

PRELIMINARY EXAMINATIONS 2026



Course 4: Mathematical Thinking (Discrete Structures)

A Definitive Mathematical Reference on Formal Logic, Proof Theory, Set Relations, Combinatorics, Graph Structures, and Number Systems for Computer Science

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Table of Contents

Module 1: Logic & Mathematical Reasoning	4
1.1 Propositions & Logical Connectives	4
1.2 Truth Tables, Tautologies & Contradictions	5
1.3 Predicates, Quantifiers & Proof Techniques	6
1.4 Principle of Mathematical Induction	8
1.5 Practice Set & Comprehensive Scenario	9
Module 2: Sets, Relations & Functions	12
2.1 Set Operations & Venn Diagram Analysis	12
2.2 Relations & Properties of Relations	13
2.3 Functions, Inverse, & Composites	15
2.4 Practice Set & Comprehensive Scenario	16
Module 3: Counting Techniques & Probability	19
3.1 Permutations, Combinations & Pigeonhole Principle	19
3.2 Inclusion-Exclusion & Basic Probability	21
3.3 Conditional Probability & Bayes' Theorem	22
3.4 Practice Set & Comprehensive Scenario	24
Module 4: Graph Theory Fundamentals	27
4.1 Graph Terminology & Vertex Degree Limits	27
4.2 Paths, Cycles, Trees & Spanning Trees	29
4.3 Graph Traversals (DFS & BFS)	30
4.4 Practice Set & Comprehensive Scenario	32
Module 5: Number Theory & Boolean Algebra	35
5.1 Divisibility, GCD & Euclidean Algorithms	35
5.2 Modular Arithmetic & Congruence Relations	37
5.3 Boolean Algebra & Logic Gate Simplification	38

5.4 Practice Set & Comprehensive Scenario 40

MODULE 1

Logic & Mathematical Reasoning

1.1 Propositions & Logical Connectives

A **Proposition** is a declarative sentence that is either strictly true (T) or false (F), but not both. For example, "Preliminary Examinations occur in 2026" is a proposition, whereas "Solve this problem!" is an imperative sentence and not a proposition.

Compound propositions are constructed by combining simple propositions using **Logical Connectives**:

- **Negation ($\neg p$)**: Flip the truth value. If p is true, $\neg p$ is false.
- **Conjunction ($p \wedge q$)**: True only if both p and q are true.
- **Disjunction ($p \vee q$)**: True if at least one of p or q is true.
- **Conditional ($p \rightarrow q$)**: False only when p (hypothesis) is true and q (conclusion) is false.
- **Biconditional ($p \leftrightarrow q$)**: True if p and q share identical truth values.

Solved Example 1.1: Connective Expressions

Let p : "It is raining" (False) and q : "The database is online" (True). Evaluate the truth value of the compound proposition: $\neg p \rightarrow q$.

Solution:

1. First, determine $\neg p$. Since p is False, $\neg p$ evaluates to True.
2. Evaluate the conditional statement $\text{True} \rightarrow \text{True}$.
3. By conditional definition, $\text{True} \rightarrow \text{True}$ is **True**.

1.2 Truth Tables, Tautologies & Contradictions

A **Truth Table** displays the output truth values of a compound proposition for all possible combinations of truth values of its constituent variables.

- **Tautology**: A compound proposition that is always true, regardless of the truth values of its individual propositions (e.g., $p \vee \neg p$).

- **Contradiction:** A compound proposition that is always false (e.g., $p \wedge \neg p$).

DE MORGAN'S LAWS (LOGICAL EQUIVALENCE)

$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$

$$\neg(p \vee q) \equiv \neg p \wedge \neg q$$

Solved Example 1.2: Proving Tautology via Truth Table

Verify if the proposition $(p \wedge (p \rightarrow q)) \rightarrow q$ is a tautology.

Solution: Construct the truth table:

p	q	$p \rightarrow q$	$p \wedge (p \rightarrow q)$	$(p \wedge (p \rightarrow q)) \rightarrow q$
T	T	T	T	T
T	F	F	F	T
F	T	T	F	T
F	F	T	F	T

Since the final column contains only "True" values, the proposition is a **Tautology** (known classically as Modus Ponens).

1.3 Predicates, Quantifiers & Proof Techniques

A **Predicate** is a statement containing variables that becomes a proposition when values are assigned to those variables (denoted $P(x)$).

Quantifiers define the scope of variables in a predicate:

- **Universal Quantifier ($\forall x$):** "For all x ," asserting that $P(x)$ is true for every element in the domain.
- **Existential Quantifier ($\exists x$):** "There exists x ," asserting that $P(x)$ is true for at least one element in the domain.

Proof Techniques

Mathematical assertions are proved using logic rules:

1. **Direct Proof:** Assume hypothesis p is true, and show that conclusion q must be true.

2. **Proof by Contrapositive:** Prove $p \rightarrow q$ by showing that if the conclusion is false ($\neg q$), then the hypothesis must be false ($\neg p$).
3. **Proof by Contradiction:** Assume the statement to be proved is false, and show that this assumption leads to a logical contradiction.

Solved Example 1.3: Proof by Contradiction

Prove that if $3n + 2$ is odd (where n is an integer), then n is odd.

Solution:

1. Assume the hypothesis is true ($3n + 2$ is odd) and the conclusion is false (n is even).
2. Since n is even, it can be written as $n = 2k$ for some integer k .
3. Substitute this into the expression: $3n + 2 = 3(2k) + 2 = 6k + 2 = 2(3k + 1)$.
4. Since $3k + 1$ is an integer, $3n + 2$ is a multiple of 2, making it even.
5. This contradicts our assumption that $3n + 2$ is odd. Therefore, n must be odd.

1.4 Principle of Mathematical Induction

Mathematical Induction is a proof technique used to prove that a statement $P(n)$ is true for all positive integers n . The proof consists of two steps:

1. **Basis Step:** Show that the base case statement $P(1)$ is true.
2. **Inductive Step:** Show that if the inductive hypothesis $P(k)$ is true for some arbitrary integer k , then the next case $P(k+1)$ must also be true.

Solved Example 1.4: Summation Induction

Prove that the sum of the first n positive integers is given by: $1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$.

Solution:

1. **Basis Step:** For $n = 1$, LHS = 1. RHS = $\frac{1(1+1)}{2} = 1$. The basis step is true.

2. **Inductive Step:** Assume the statement is true for $n = k$:

$$1 + 2 + 3 + \dots + k = \frac{k(k+1)}{2}.$$

3. Show it is true for $n = k + 1$:

$$\text{LHS} = (1 + 2 + 3 + \dots + k) + (k + 1) = \frac{k(k+1)}{2} + (k + 1).$$

4. Factor out $(k + 1)$:

$$\text{LHS} = (k + 1) \left(\frac{k}{2} + 1 \right) = \frac{(k + 1)(k + 2)}{2} = \text{RHS for } n = k + 1.$$

By induction, the statement is true for all positive integers n .

Module 1 - Practice Exercise Set

1 Logical Connective Translation

Let p : "The compiler compiles", q : "The database is locked". Translate the statement into symbolic logic: "If the compiler compiles, then the database is locked or it is not the case that the compiler compiles".

2

Evaluating Tautology

Use a truth table to determine if the proposition $(p \rightarrow q) \wedge (q \rightarrow p)$ is a tautology, contradiction, or contingency.

3

Quantifier Scope Logic

Let the domain of discourse be all integers. Evaluate the truth value of: $\forall x \exists y (x + y = 0)$. Explain your reasoning.

4

Contrapositive Proof Verification

Write the contrapositive of the statement: "If n^2 is even, then n is even". Prove the contrapositive.

5

Inductive Inequality

Use mathematical induction to prove that $2^n > n$ for all positive integers n .

6

Equivalence Verification

Prove the logical equivalence: $\neg(p \rightarrow q) \equiv p \wedge \neg q$ using truth tables.

7

Negating Quantified Statements

Find the negation of the quantified statement: $\exists x \forall y (P(x, y) \rightarrow Q(x, y))$.

8

Direct Proof Check

Prove directly that the product of two odd integers is odd.

9

Contrapositive Structure

Prove that if n is an integer and $n^3 + 5$ is odd, then n is even.

10**Induction: Sum of Squares**

Prove by induction that $1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$.

Module 1 - Comprehensive Problem

Scenario: Automated Logical Verification System

A safety-critical automated train control loop depends on three propositional sensors: p ("Train is on track"), q ("Signal light is green"), and r ("Manual override active"). The system operates under the safety rule: "If the train is on track, then manual override must be active or the signal light must be green."

- **(a)** Write the safety rule as a compound propositional logic statement.
- **(b)** Construct the full truth table for the safety rule proposition.
- **(c)** Determine under which sensor input combination the safety rule evaluates to False.
- **(d)** Find the logical negation of the safety rule. Simplify it using De Morgan's laws.
- **(e)** If the manual override sensor fails (r is always False), prove by contradiction that if the safety rule is True and the train is on track, the signal light must be green.

MODULE 2

Sets, Relations & Functions

2.1 Set Operations & Venn Diagram Analysis

A **Set** is an unordered collection of distinct objects. A subset $A \subseteq B$ means every element of A is also an element of B . The **Power Set** $P(A)$ is the set of all subsets of A . If $|A| = n$, then $|P(A)| = 2^n$.

Set operations are defined as:

- **Union** ($A \cup B$): $\{x \mid x \in A \vee x \in B\}$.
- **Intersection** ($A \cap B$): $\{x \mid x \in A \wedge x \in B\}$.
- **Difference** ($A \setminus B$): $\{x \mid x \in A \wedge x \notin B\}$.
- **Cartesian Product** ($A \times B$): $\{(a, b) \mid a \in A \wedge b \in B\}$.

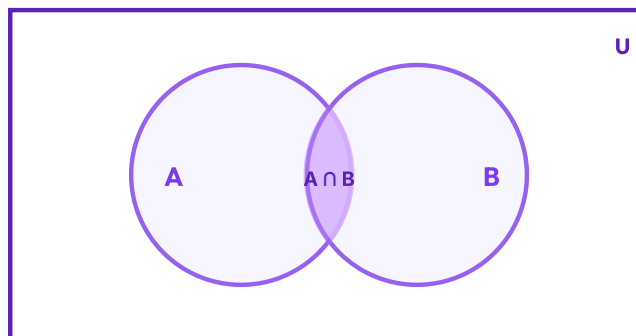


Figure 2.1: Venn diagram showing the intersection subset of two overlapping sets.

Solved Example 2.1: Power Set Generation

Given the set $S = \{a, b\}$. Find the power set $P(S)$ and verify its cardinality.

Solution:

1. The subsets of S are:
 - The empty set: $\emptyset$
 - Single-element subsets: $\{a\}, \{b\}$
 - The set itself: $\{a, b\}$
2. Therefore, $P(S) = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$.
3. Cardinality check: $|S| = 2$. $|P(S)| = 2^2 = 4$, which matches our list.

2.2 Relations & Properties of Relations

A binary **Relation** R from set A to set B is a subset of the Cartesian product $A \times B$. If $(a, b) \in R$, we write aRb .

A relation R on a single set A can have several key properties:

- **Reflexive:** $\forall a \in A, (a, a) \in R$.
- **Symmetric:** $\forall a, b \in A, (a, b) \in R \rightarrow (b, a) \in R$.
- **Transitive:** $\forall a, b, c \in A, ((a, b) \in R \wedge (b, c) \in R) \rightarrow (a, c) \in R$.

An **Equivalence Relation** is a relation that is reflexive, symmetric, and transitive.

Solved Example 2.2: Relation Classification

Let $A = \{1, 2, 3\}$. Consider the relation $R = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 1)\}$. Determine if R is an equivalence relation.

Solution:

1. **Reflexive:** Yes, because $(1, 1)$, $(2, 2)$, and $(3, 3)$ are all in R .
2. **Symmetric:** Yes, because $(1, 2) \in R$ and its inverse $(2, 1) \in R$.
3. **Transitive:** Yes. The only non-trivial chains to check are:
 - $(1, 2) \in R$ and $(2, 1) \in R \rightarrow (1, 1) \in R$ (True).
 - $(2, 1) \in R$ and $(1, 2) \in R \rightarrow (2, 2) \in R$ (True).

Since R is reflexive, symmetric, and transitive, it is an **Equivalence Relation**.

2.3 Functions, Inverse, & Composites

A **Function** $f: A \rightarrow B$ is a relation that assigns each element in set A (domain) to exactly one element in set B (codomain).

- **One-to-One (Injective):** $\forall x_1, x_2 \in A, f(x_1) = f(x_2) \rightarrow x_1 = x_2$.
- **Onto (Surjective):** $\forall y \in B, \exists x \in A$ such that $f(x) = y$.
- **Bijection:** A function that is both injective and surjective. Only bijective functions have an **Inverse Function** f^{-1} .

FUNCTION COMPOSITION

Given $f: A \rightarrow B$ and $g: B \rightarrow C$

The composition $(g \circ f): A \rightarrow C$ is defined as:

$$(g \circ f)(x) = g(f(x))$$

Solved Example 2.3: Function Composition

Let $f(x) = x + 3$ and $g(x) = x^2$ be functions from $\mathbb{R}$ to $\mathbb{R}$. Find the compositions $(g \circ f)(x)$ and $(f \circ g)(x)$.

Solution:

$$1. (g \circ f)(x) = g(f(x)) = g(x + 3) = (x + 3)^2 = x^2 + 6x + 9.$$

$$2. (f \circ g)(x) = f(g(x)) = f(x^2) = x^2 + 3.$$

Note that $(g \circ f)(x) \neq (f \circ g)(x)$, showing that function composition is non-commutative.

Module 2 - Practice Exercise Set

1

Venn Diagram Verification

Let $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5, 6\}$, and $C = \{4, 6, 7, 8\}$. Find $(A \cup B) \cap C$.

2

Power Set Identification

Find the power set $P(A)$ for $A = \{1, \{2\}\}$.

3

Cartesian Product Cardinality

If $|A| = 4$ and $|B| = 3$, find the cardinality of $P(A \times B)$.

4

Relation Property Check

Let R be a relation on the set of integers such that $(a, b) \in R$ if and only if $a - b$ is even. Prove that R is reflexive.

5

Transitive Property Verification

Determine if the relation $R = \{(1, 2), (2, 3)\}$ on set $S = \{1, 2, 3\}$ is transitive. If not, find the smallest transitive closure.

6

Injective Function Proof

Prove that the function $f(x) = 2x + 5$ from $\mathbb{R}$ to $\mathbb{R}$ is one-to-one.

7

Onto Function Verification

Determine if the function $f(n) = n + 1$ from the domain of integers to integers is onto.

8

Inverse Function Calculation

Find the inverse $f^{-1}(x)$ for the bijective function $f(x) = \frac{x + 2}{3}$.

9

Composite Function Chain

Given $f(x) = x^2$ and $g(x) = 2x$, find the value of $(g \circ f)(3)$.

10

Set Difference Equivalence

Prove algebraically that $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$.

Module 2 - Comprehensive Problem

Scenario: Relational Database Partitioning

A relational database table holds user account rows. Let U be the set of all user accounts. We define a binary relation R on U such that $u_1 R u_2$ if and only if u_1 and u_2 were registered in the same month.

- **(a)** Prove that R is an equivalence relation.
- **(b)** Explain what partition equivalence classes U/R represent in the context of database partitioning.
- **(c)** Let $f: U \rightarrow \mathbb{N}$ be a function mapping a user account to their birth year. If $U = \{u_1, u_2, u_3\}$ and their birth years are 1995, 2000, and 1995 respectively, determine if f is injective, surjective, or bijective.
- **(d)** Let $g: \mathbb{N} \rightarrow \mathbb{N}$ be a function defined as $g(x) = x - 1900$. Write the formula for the composite function $(g \circ f)(u)$.
- **(e)** Find the inverse function $(g \circ f)^{-1}$ if the domain of U is restricted to $\{u_1, u_2\}$ and their corresponding birth years are $\{1995, 2000\}$.

MODULE 3

Counting Techniques & Probability

3.1 Permutations, Combinations & Pigeonhole Principle

Counting techniques allow us to determine the number of possible outcomes in a sample space without listing them individually:

- **Sum Rule:** If a task can be done in m ways and a second task in n ways, and the tasks cannot be done simultaneously, then there are $m + n$ ways to do one of the tasks.
- **Product Rule:** If a task consists of two successive steps, where step 1 can be done in m ways and step 2 in n ways, then there are $m \times n$ ways to complete the task.

PERMUTATIONS VS. COMBINATIONS FORMULAS

Permutations (Order matters):

$$P(n, r) = \frac{n!}{(n - r)!}$$

Combinations (Order does not matter):

$$C(n, r) = \frac{n!}{r!(n - r)!}$$

Pigeonhole Principle: If $k + 1$ or more objects are placed into k boxes, then at least one box must contain two or more objects.

Solved Example 3.1: Password Combinations

A system password consists of 3 letters followed by 2 digits. How many unique passwords can be generated if repetition is allowed?

Solution:

1. By product rule, analyze the choices for each character:
 - Each of the 3 letters has 26 possibilities.
 - Each of the 2 digits has 10 possibilities (0-9).
2. Multiply the choices together:

$$N = 26 \times 26 \times 26 \times 10 \times 10 = 26^3 \times 10^2 = 1,757,600 \text{ unique passwords.}$$

3.2 Inclusion-Exclusion & Basic Probability

The **Principle of Inclusion-Exclusion** calculates the size of the union of multiple sets by accounting for elements counted twice in the intersection:

INCLUSION-EXCLUSION FORMULA

$$|A \cup B| = |A| + |B| - |A \cap B|$$

Probability: The probability of an event E occurring within a finite, equally likely sample space S is defined as:

$$P(E) = \frac{|E|}{|S|}.$$

Solved Example 3.2: Inclusion-Exclusion

In a class of 50 students, 30 study Python, 25 study C++, and 10 study both. How many students study neither language?

Solution:

1. Let P be the set of Python students ($|P|=30$) and C be the set of C++ students ($|C|=25$).
2. The students studying at least one language is:

$$|P \cup C| = |P| + |C| - |P \cap C| = 30 + 25 - 10 = 45 \text{ students.}$$
3. Students studying neither: $50 - 45 = 5$ students.

3.3 Conditional Probability & Bayes' Theorem

The **Conditional Probability** of event A given that event B has already occurred is:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

Bayes' Theorem: Relates conditional probabilities and allows us to update the probability of a hypothesis as new evidence becomes available:

BAYES' THEOREM FORMULA

$$P(A \mid B) = \frac{P(B \mid A) P(A)}{P(B)}$$

Solved Example 3.3: Bayes' Theorem

A medical test is 99% accurate for detecting a disease that affects 1% of the population. If a patient tests positive, what is the probability that they actually have the disease?

Solution:

1. Let D : Patient has disease ($P(D) = 0.01$). Let T : Patient tests positive.
2. We are given the test accuracy: $P(T \mid D) = 0.99$ and the false positive rate: $P(T \mid \neg D) = 0.01$.
3. Find the total probability of testing positive $P(T)$:

$$P(T) = P(T \mid D)P(D) + P(T \mid \neg D)P(\neg D) = (0.99 \times 0.01) + (0.01 \times 0.99) = 0.0198.$$

4. Apply Bayes' Theorem:

$$P(D \mid T) = \frac{P(T \mid D)P(D)}{P(T)} = \frac{0.99 \times 0.01}{0.0198} = 0.50 \text{ (50\%).}$$

Module 3 - Practice Exercise Set

1

Product Rule Routing

A traveler can route from City A to City B in 4 ways, and from City B to City C in 5 ways. How many unique routes exist from A to C through B?

2

Permutations Calculation

How many unique 4-letter words can be formed from the letters of the word "COMPUTER" if repetition is not allowed?

3

Combinations Selection

A committee of 3 programmers is to be chosen from a pool of 8. In how many ways can this committee be selected?

4

Pigeonhole Proof

Prove that if 8 integers are chosen from the set $\{1, 2, \dots, 14\}$, at least one pair must sum to 15.

5

Inclusion-Exclusion Triple Sets

State the Inclusion-Exclusion formula for three sets: $|A \cup B \cup C|$.

6

Dice Roll Probability

Two fair six-sided dice are rolled. Find the probability that the sum of the numbers shown is 8.

7

Conditional Probability Cards

A card is drawn from a standard deck. What is the probability that the card is a King, given that it is a Face card?

8

Indistinguishable Permutations

Find the number of unique arrangements of the letters in the word "DATABASE".

9

Pigeonhole Application

What is the minimum number of students needed in a class to guarantee that at least two students share the same birth month?

10

Conditional Event Multiplication

If $P(A) = 0.6$, $P(B) = 0.4$, and $P(A \cap B) = 0.2$, calculate the conditional probability $P(B \mid A)$.

Module 3 - Comprehensive Problem

Scenario: Network Packet Loss Diagnosis

An IT network routing link receives packets from three servers: S_1 (sends 50% of traffic), S_2 (sends 30%), and S_3 (sends 20%). The probability of packet corruption/loss for each server is: 1% for S_1 , 2% for S_2 , and 5% for S_3 .

- **(a)** Calculate the probability that a packet is from S_1 and is corrupted.
- **(b)** Calculate the probability that a packet is from S_2 and is corrupted.
- **(c)** Find the total probability that a packet received on the link is corrupted.
- **(d)** If a packet is found to be corrupted, use Bayes' Theorem to calculate the probability that it was sent by S_3 .
- **(e)** If the system administrator wants to minimize overall packet corruption by routing traffic away from the worst link, which server should be prioritized for hardware replacement? Explain using the probabilities calculated.

MODULE 4

Graph Theory Fundamentals

4.1 Graph Terminology & Vertex Degree Limits

A **Graph** $G = (V, E)$ consists of a set of vertices V (nodes) and a set of edges E (connections between vertices). Graphs can be directed (edges have arrows indicating direction) or undirected.

The **Degree** of a vertex v , denoted $\deg(v)$, is the number of edges incident to it. In a directed graph, we differentiate between in-degree (incoming edges) and out-degree (outgoing edges).

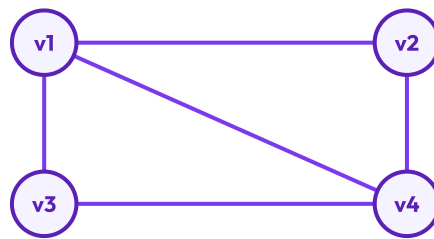


Figure 4.1: A simple undirected graph containing 4 vertices and 5 edges.

THE HANDSHAKING LEMMA

For any undirected graph $G = (V, E)$:

$$\sum_{v \in V} \deg(v) = 2|E|$$

Solved Example 4.1: Applying Handshaking Lemma

An undirected graph has 6 vertices. If every vertex has a degree of 4, how many edges does the graph have?

Solution:

1. Calculate the sum of the degrees: $\sum \deg(v) = 6 \times 4 = 24$.
2. Apply the Handshaking Lemma: $2|E| = 24$.
3. Solve for $|E|$: $|E| = 12$ edges.

4.2 Paths, Cycles, Trees & Spanning Trees

Path: A sequence of vertices where each adjacent pair is connected by an edge. A path is simple if it does not visit any vertex more than once.

Cycle: A path that starts and ends at the same vertex, containing no other repeated vertices.

Tree: A connected, acyclic undirected graph. A tree containing n vertices has exactly $n - 1$ edges.

Spanning Tree: A subgraph of a connected graph G that contains all vertices of G and is a tree.

Solved Example 4.2: Spanning Tree Count

For a complete graph K_4 with 4 vertices, calculate the total number of unique spanning trees.

Solution:

1. Apply Cayley's Formula for complete graphs: $T(n) = n^{n-2}$ spanning trees.
2. For $n = 4$: $T(4) = 4^{4-2} = 4^2 = 16$ spanning trees.

4.3 Graph Traversals: DFS & BFS

Graph traversal algorithms visit all vertices of a graph systematically:

- **Depth-First Search (DFS):** Starts at a source vertex and explores as far as possible along each branch before backtracking. It uses a **Stack** data structure (often recursively).

- **Breadth-First Search (BFS):** Starts at a source vertex and explores all neighboring vertices at the current depth level before moving to vertices at the next depth level. It uses a **Queue** data structure.

Solved Example 4.3: BFS Traversal Trace

Trace a BFS traversal on the graph in Figure 4.1 starting at vertex v_1 . Assume adjacency lists are in alphabetical order.

Solution:

1. Initialize queue $Q = [v_1]$. Visited list = $\{v_1\}$.
2. Dequeue v_1 . Check neighbors of v_1 : v_2, v_3, v_4 . Queue all unvisited neighbors: $Q = [v_2, v_3, v_4]$. Visited list = $\{v_1, v_2, v_3, v_4\}$. Output: v_1 .
3. Dequeue v_2 . Neighbors of v_2 are v_1, v_4 . Both are already visited. Queue: $Q = [v_3, v_4]$. Output: v_1, v_2 .
4. Dequeue v_3 . Neighbors of v_3 are v_1, v_4 . Both are already visited. Queue: $Q = [v_4]$. Output: v_1, v_2, v_3 .
5. Dequeue v_4 . Queue is empty. Output: v_1, v_2, v_3, v_4 .

The BFS traversal order is: v_1, v_2, v_3, v_4 .

Module 4 - Practice Exercise Set

1

Sum of Degrees

A graph contains 8 edges. Calculate the sum of the degrees of all its vertices.

2

Graph Feasibility Check

Determine if an undirected graph can exist with vertex degrees $\{1, 2, 2, 3, 5\}$. Explain your reasoning.

3

Complete Graph Edges

Calculate the number of edges in a complete graph K_5 with 5 vertices.

4

Tree Properties

A tree has 15 vertices. How many edges does it contain?

5

Cayley's Tree Count

Calculate the number of unique spanning trees for a complete graph with 3 vertices.

6

Bipartite Graph Definition

Define a bipartite graph. Is a cycle graph C_4 bipartite?

7

DFS Traversal Queue

Name the primary data structure used to implement DFS. Explain why it is suitable.

8

Adjacency Matrix Mapping

Draw the adjacency matrix representation for the graph in Figure 4.1.

9

Path vs. Cycle

Explain the difference between a path and a cycle in graph theory.

10

Isomorphism Check

Explain the concept of graph isomorphism. What invariant properties must isomorphic graphs share?

Module 4 - Comprehensive Problem

Scenario: Network Routing Backplane Topology

A cloud server room contains 5 switch routers (A, B, C, D, E). The link configurations between routers are: $A-B$, $A-C$, $B-C$, $B-D$, $C-D$, and $D-E$.

- **(a)** Sketch the graph representation of this network. List the vertex degrees for all 5 routers.
- **(b)** Verify the Handshaking Lemma for this network graph.
- **(c)** Trace a DFS traversal of the network starting from router A (assume alphabetical selection for adjacent paths).
- **(d)** Find a spanning tree for this network using Kruskal's or Prim's algorithm, assuming all links have equal weight.
- **(e)** If router D fails (removing all its connections), is the remaining network graph still connected? List the connected components.

MODULE 5

Number Theory & Boolean Algebra

5.1 Divisibility, GCD & Euclidean Algorithms

Number Theory is the study of integers and their properties:

- **Divisibility:** An integer a divides an integer b (written $a \mid b$) if there exists an integer c such that $b = ac$.
- **Greatest Common Divisor (GCD):** The largest positive integer that divides both a and b (denoted $\gcd(a, b)$).

The **Euclidean Algorithm** is an efficient method for computing the GCD of two integers. It is based on the principle that the GCD of two numbers also divides their difference:
 $\gcd(a, b) = \gcd(b, a \bmod b)$.

Solved Example 5.1: Euclidean Algorithm GCD

Find the greatest common divisor $\gcd(252, 105)$ using the Euclidean Algorithm.

Solution: Perform successive modulo divisions:

1. $252 = 2 \times 105 + 42 \rightarrow \gcd(252, 105) = \gcd(105, 42)$
2. $105 = 2 \times 42 + 21 \rightarrow \gcd(105, 42) = \gcd(42, 21)$
3. $42 = 2 \times 21 + 0 \rightarrow \gcd(42, 21) = 21$.

The last non-zero remainder is 21. Therefore, $\gcd(252, 105) = 21$.

5.2 Modular Arithmetic & Congruence Relations

If a and b are integers and m is a positive integer, then a is **congruent to b modulo m** if m divides $a - b$. We write this relation as:

$$a \equiv b \pmod{m}.$$

Modular arithmetic is fundamental to cryptography algorithms like RSA.

Solved Example 5.2: Solving Linear Congruence

Find all integer solutions x to the linear congruence: $3x \equiv 2 \pmod{5}$.

Solution:

1. To solve $3x \equiv 2 \pmod{5}$, we need to find the modular inverse of 3 modulo 5.
2. We test values for y in $3y \equiv 1 \pmod{5}$:
 - $3 \times 1 = 3 \equiv 3$
 - $3 \times 2 = 6 \equiv 1 \pmod{5}$.

Thus, the modular inverse of 3 modulo 5 is 2.

3. Multiply both sides of the congruence by 2:
 $2 \times (3x) \equiv 2 \times 2 \pmod{5} \rightarrow x \equiv 4 \pmod{5}$.

The integer solutions are $x = 4 + 5k$ for any integer k .

5.3 Boolean Algebra & Logic Gate Simplification

Boolean Algebra deals with variables that have two discrete values: 1 (True) and 0 (False). The primary Boolean operations are:

- **AND (Multiplication, $x \cdot y$):** Outputs 1 only if both inputs are 1.
- **OR (Addition, $x + y$):** Outputs 1 if at least one input is 1.
- **NOT (Inversion, $\bar{x}$ or x'):** Flips the input value.



Figure 5.2: Schematic representations of the AND logic gate and NOT inverter gate.

Solved Example 5.3: Boolean Simplification

Simplify the Boolean expression using algebraic laws: $F = AB'C + AC$.

Solution:

1. Factor out the common term AC :

$$F = AC(B' + 1).$$

2. Apply the Annulment Law ($B' + 1 = 1$):

$$F = AC(1).$$

3. Apply the Identity Law ($AC \cdot 1 = AC$):

$$F = AC.$$

The simplified expression is: $F = AC$.

Module 5 - Practice Exercise Set

1

Divisibility proof

Prove that if $a \mid b$ and $b \mid c$, then $a \mid c$ for any integers a, b, c .

2

Greatest Common Divisor

Find the greatest common divisor $\gcd(120, 500)$.

3

Euclidean trace

Use the Euclidean algorithm to find the value of $\gcd(412, 190)$. Show all modulo division steps.

4

Modular Arithmetic Calculation

Find the remainder when 3^{10} is divided by 5.

5

Linear Congruence Equation

Solve the congruence equation $2x \equiv 3 \pmod{7}$.

6

Binary Addition

Perform binary addition on the numbers 1101_2 and 1011_2 . Convert the result to decimal to verify.

7

Boolean Inversion

Apply De Morgan's laws to simplify the Boolean expression: $A \cdot (B + C')$ with an overall inversion line.

8

Prime Decomposition

Find the prime factorizations of 140 and 250, and use them to calculate $\text{lcm}(140, 250)$.

9

Euclidean modular inverse

Find the modular multiplicative inverse of 7 modulo 26.

10

Boolean Distributive Law

Prove the Boolean distributive law: $A + (B \cdot C) = (A + B) \cdot (A + C)$ using a truth table.

Module 5 - Comprehensive Problem

Scenario: Cryptographic Key Generator Logic

An encryption key generation chip uses modular arithmetic and logic gate circuits to verify key integrity. The chip generates public/private key pairs using the modulus $n = pq$, where $p = 11$ and $q = 13$ are prime numbers.

- **(a)** Calculate the modulus n and the totient function $\varphi(n) = (p-1)(q-1)$.
- **(b)** If the public exponent is chosen as $e = 7$, prove that $\gcd(e, \varphi(n)) = 1$ using the Euclidean Algorithm.
- **(c)** Calculate the private decryption key d , which is the modular inverse of e modulo $\varphi(n)$ (i.e., $7d \equiv 1 \pmod{\varphi(n)}$).
- **(d)** The hardware chip contains a Boolean logic validator circuit that outputs a check bit $F = AB + A(B + C) + B(B + C)$. Simplify the expression algebraically.
- **(e)** Write down the simplified logic expression for the validator circuit.